Exact Resonance Laws for Compressed Cat Maps: Matched Identity, Ensemble Mean, and Perturbation-Induced Wigner–Dyson Statistics

August 2026

Source edit log (2026-08-28): the `.tex` source was edited on a machine with *no* LaTeX engine; the last successful `.pdf` compile is the 159.4 KB file dated 2026-08-27. Edits applied in this source revision (without recompile) are: (1) the $\gcd(M_{21},\ N)=1$ hypothesis on the anti-unitary $\Theta^2=+1$ result (Remark 5, and §8 status table footnote); (2) the `lem:fiber` non-degeneracy hypothesis (a_1,a_3,b_1,b_3 all nonzero, footnote in §4); (3) the corrected U4 bound EXTRAS $\le k^3 p^{-2\delta}$ with the explicit $k \ll p^{2\delta}$ threshold (§4, Tail subsection); (4) the r=5 master-formula demoted to "numerically supported for $r \le 4$; one pass at $r=5$ (Arnold @ $N=101$, err$\le 4\times10^{-16}$); legacy `m1_new_configs.py` and `s_of_j5.jl` deprecated" (§8 status table); (5) a new T21 status row recording that the $k \ge p^{3/7+\varepsilon}$ smooth law is unchanged as of 2026-08-28, with the three Ring 2 attack lines (Stepanov / Deligne / Bourgain) all closing as [OPEN GAP]. **Do not re-pdf-compile this source without auditing every edit against the LEDGER RUN 64–66 entries and the `artifacts/ring2/synthesis.md`.**

Abstract

We study spectral statistics of compressions $V_\lambda = P_\lambda V P_\lambda$ of observables in degenerate eigenspaces of quantised hyperbolic toral automorphisms (Arnold cat maps). Three quantitative results are obtained for the arithmetic resonance counts that govern the moments of these compressions.

(i) Matched identity. For fully matched labels the fourth-order resonance count equals exactly $2k^2-k$ when k is odd and $3k(k-1)$ when k is even, where k is the order of the split eigenvalue α generating $H=\langle\alpha\rangle\subseteq\mathbb{F}_p^\times$. These are precisely the Wick-pairing masses for GUE and GOE; the third pairing in the even case is an arithmetic consequence of $-1\in H$, not of any classical symmetry. Classical reversibility of the map does not force k to be even.

(ii) Ensemble-mean law. Averaged over the *full* label ensemble $(a,b)\in(\mathbb{F}_p^\times)^8$, the reduced resonance count has an *exactly computable* mean, $\langle N_3\rangle = k^3(p^2-3p+3)^2/(p-1)^6 = k^3/p^2\,(1+O(1/p))$, proved by elementary double counting. Degenerate patterns form a set of relative measure $O(1/p)$, so the conditional mean over generic patterns shares the same leading term: a typical pattern has small expected resonance count once $k<p^{2/3}$, and nonzero counts are rare fluctuations around this small mean. A companion tail bound $\max N_3 \le \lfloor k^2/p\rfloor+2$ over generic patterns is supported by zero violations across ∼230 configurations.

(iii) Perturbation-induced Wigner–Dyson statistics. Under the unitary perturbation flow $U_\varepsilon = e^{i\varepsilon E}U_A$ with E drawn from the Gaussian unitary ensemble, intra-multiplet level spacings — tracked through the flow by optimal assignment — agree with the Wigner–Dyson surmise and clearly reject Poisson across four independent primes ($p\in\{1597,\dots,3001\}$, up to 204 spacings per prime). Scaling the same pipeline to three further primes (13 723 spacings, reference ensembles matched to the observed multiplet dimensions) resolves the ensemble class: the data match GUE and exclude GOE with a KS gap five times the critical value, exactly as Lemma 18

predicts for a GUE perturbation. This links the exact arithmetic core to random-matrix phenomenology under an infinitesimal symmetry breaking; it does not establish an unconditional local spacing law for the unperturbed compressions, which remains open.

Contents

1 Introduction **2**

2 Setup **3**
2.1 Classical and quantised dynamics 3
2.2 Resonance counts 4

3 The matched sector **4**

4 The unmatched sector **6**
4.1 Reduction to H^3 6
4.2 Tail: the tight resonance bound 6
4.3 Mean: an exact ensemble law 7
4.4 Complete picture of the fourth moment 8

5 The sixth moment: strata laws and an arithmetic excess **9**

6 Perturbation-induced universality **10**

7 What remains open **11**

8 Summary of statuses **12**

A Methodological details **12**

1 Introduction

The Bohigas–Giannoni–Schmit (BGS) conjecture asserts that quantum spectra of classically chaotic systems exhibit universal fluctuations described by random-matrix ensembles [1]: GOE for generic time-reversal-invariant systems, GUE when time reversal is absent, GSE with spin-1/2 and time reversal.

Quantised Arnold cat maps — hyperbolic elements of $\mathrm{SL}(2,\mathbb{Z})$ acting on the torus, quantised via Weil's metaplectic representation — provide one of the cleanest laboratories for such questions [2]. Kurlberg–Rudnick [3, 4] showed that the Hecke algebra diagonalises U_A and that matrix coefficients of Hecke eigenfunctions become Gaussian in the semiclassical limit. Raw (non-Hecke-symmetrised) maps, however, exhibit large eigenspace multiplicities: if $\alpha \in \mathbb{F}_p^\times$ is a split eigenvalue of A mod p and $H = \langle \alpha \rangle$ with $k = |H|$, every eigenvalue of U_A has multiplicity at least $\lfloor p/K \rfloor$, where K is the quantum period [3]. Inside one eigenspace the compression $V_\lambda := P_\lambda V P_\lambda$ of an observable V is an $m \times m$ matrix whose statistics invite comparison with random-matrix predictions.

Moments of V_λ reduce to *resonance counts*: solutions in H^r of two coupled linear equations, one linear in the variables h_i, one in their reciprocals. An earlier draft of this programme claimed a "fine-scale Wigner–Dyson theorem" at order four with error $C_F k^8/p^2$ valid for $k = o(p^{2/3})$. *[Historical note: that argument double-counted a prefactor between two equivalent formulations of the resonance count and was retracted; it is mentioned here only to record provenance, and no*

threshold of this shape is used or needed below.] The present paper replaces asymptotic claims by exact statements and carefully separated statistical levels.

Contributions

1. **Matched identity (Theorem 2).** A complete proof, by structural decomposition into pairing types plus an $S = 0$ layer, that the fully matched fourth-order count equals $2k^2 - k$ (k odd) resp. $3k(k-1)$ (k even). Verified numerically in dozens of configurations and manually at $k = 2, 4$.

2. **Reversibility vs. parity (Remark 5).** Numerous reversible maps admit split primes with odd k; the GUE-type value of the counter there is forced by arithmetic alone.

3. **Ensemble-mean law (Theorem 12).** An exact closed form for the mean unmatched resonance count over random labels, proved by double counting together with new zero-count lemmas for linear forms on $(\mathbb{F}_p^\times)^n$ (Lemma 11); confirmed within Monte-Carlo errors in five configurations ($|z| < 1.5$).

4. **Tail bound (Conjecture 7).** The empirical law $\max_{a,b} N_3 \le \lfloor k^2/p \rfloor + 2$ for generic patterns, with zero violations across ∼230 tested configurations; distinguished from the mean and from degenerate sectors, which obey their own closed forms.

5. **Perturbation-induced universality (Lemma 18, Experiment 19).** Under the unitary flow $U_\varepsilon = e^{i\varepsilon E} U_A$ with E GUE, intra-multiplet splittings follow a GUE compression law; tracked spacings follow Wigner–Dyson and reject Poisson across four primes.

Throughout, status tags distinguish proven statements from numerically supported ones (Section 8); no BGS claim is made beyond what these tags license.

2 Setup

2.1 Classical and quantised dynamics

Let $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}(2,\mathbb{Z})$ be hyperbolic, $|\operatorname{tr} A| > 2$, and let $N = p$ be an odd prime in the *split* regime: $\Delta := (\operatorname{tr} A)^2 - 4 \not\equiv 0$ is a quadratic residue mod p, so A has eigenvalues $\alpha, \alpha^{-1} \in \mathbb{F}_p^\times$. The inert case lies outside this paper (Open Problem 24).

On $\mathcal{H}_p = L^2(\mathbb{Z}/p)$, with $\omega = e^{2\pi i/p}$ and $b_2 = 2^{-1} \bmod p$, define the Weyl operators

$$T(n_1, n_2) = \omega^{b_2 n_1 n_2}\, Q(n_2) P(n_1), \qquad (Q(n)\psi)(x) = \omega^{nx}\psi(x), \quad (P(n)\psi)(x) = \psi(x-n),$$

satisfying $T(g)T(h) = \omega^{\Psi(g,h)/2} T(g+h)$ with $\Psi(g,h) = g_2 h_1 - g_1 h_2$. The Weil quantisation U_A is fixed (up to phase) by Egorov $U_A^{-1} T(v) U_A = T(Av)$ and obeys $U_A^K = \chi(A)\, I$, where the *quantum period*

$$K \;:=\; \operatorname{ord}_{\mathrm{PSU}(2,\mathbb{Z}/p)}(A)$$

is the order of the image of A in $\mathrm{PSU}(2,\mathbb{Z}/p)$ (cf. [5]) and $\chi(A) = \pm 1$ is the metaplectic sign. Eigenspaces of U_A itself are the objects of study.

For later use we record the machine-level bridge between moments and resonance counts: expanding $\operatorname{Tr}(V_\lambda^r)$ through the Weyl product rule produces, for each label pattern, the count $\mathrm{sol}(a,b)$ multiplied by an explicit unit-modulus phase and the coefficients c_n; for $r \le 4$ this "master formula"

is certified to machine precision (relative error $\leq 10^{-14}$ over a regression suite), so bounding counts bounds moment contributions in absolute value regardless of phases.

Fix a split eigenvalue λ, let $\mathcal{H}_\lambda = \ker(U_A - \lambda I)$ have dimension m with projector P_λ, and let

$$V = \sum_{n \in F} c_n T(n)$$

be an observable with finite frequency support F; write $V_\lambda = P_\lambda V P_\lambda$. Since the classical map is ergodic, the spectrum of V_λ concentrates near $\mu = m^{-1}\operatorname{Tr}(V_\lambda)$; fine-scale statistics concern the centred operator $W := V_\lambda - \mu \mathbf{1}$.

2.2 Resonance counts

Traces $\operatorname{Tr}(V_\lambda^r)$ expand over ordered signed frequency tuples; the coefficient of a pattern $(a_i; b_i) \in (\mathbb{F}_p^\times)^8$ with signs $(+,-,+,-)$ involves the number of tuples $(h_1, \ldots, h_r) \in H^r$ solving

$$a_1 h_1 - a_2 h_2 + a_3 h_3 - a_4 h_4 = 0, \qquad b_1 h_1^{-1} - b_2 h_2^{-1} + b_3 h_3^{-1} - b_4 h_4^{-1} = 0 \pmod{p}, \tag{1}$$

generalised to r slots in the obvious way. We write $\operatorname{sol}(a, b)$ for the count and reserve $\operatorname{sol}_{+-+-}(H)$ for the fully matched pattern $a_i = b_i = 1$.

Remark 1 (Collision lemma)**.** *For a slot-pair with labels $(a_1, a_3; b_1, b_3)$ define $\Phi_{13} : H^2 \to G^2$, $\Phi_{13}(x, y) = \big(a_1 x + a_3 y,\ b_1/x + b_3/y\big)$, and likewise Φ_{24}.* Non-degeneracy hypothesis (added 2026-08-28): *assume*

$$a_1 \neq 0,\ a_3 \neq 0,\ b_1 \neq 0,\ b_3 \neq 0, \qquad \text{and} \qquad a_1 b_3 \neq a_3 b_1.$$

(the first four conditions force the chart to be non-degenerate; the last forces the non-proportionality needed for the quadratic fibre argument). Under this hypothesis, every fibre of Φ_{13} has at most two points: substituting $y = (u - a_1 x)/a_3$ into the cleared reciprocal equation gives a quadratic in x. Consequently a resonance value (u, v) has at most $|\Phi_{13}^{-1}| \cdot |\Phi_{24}^{-1}| \leq 4$ preimage tuples; this is the collision lemma *$C(u, v) \leq 4$. Its algebraic core — two pairs over a field sharing both sum and product coincide up to order — is formalised in Lean 4 (`BgsFormal.lean`, theorems `prod_eq`, `sum_sq`; repository path recorded in the artefact manifest). Degenerate (proportional) slot-pairs or charts with one of a_1, a_3, b_1, b_3 equal to zero* escape *the quadratic bound: in the red-team scan on the chart $(a_1, a_3) = (0, 0)$ with $H = \langle \alpha \rangle$ of order k, the fibre of Φ_{13} is the whole H^2 of size k^2; the degenerate-stratum contribution is bounded by the explicit* strata *law (Theorem 12 and the R3 audit `audit_degenerate_strata.json`): the degenerate strata form a set of relative measure $O(1/p)$ in the label ensemble, and the leading-order contribution of degenerate patterns is captured by the mean law $k^3(p^2 - 3p + 3)^2/(p-1)^6$. Degenerate patterns are excluded from the U1 statement by the non-degeneracy hypothesis above, and are treated separately via the mean law wherever they occur.*

3 The matched sector

Theorem 2 (Exact resonance core)**.** *For any subgroup $H \subseteq \mathbb{F}_p^\times$, $k = |H|$,*

$$\operatorname{sol}_{+-+-}(H) = \begin{cases} 2k^2 - k, & k \text{ odd}, \\ 3k(k-1), & k \text{ even}. \end{cases}$$

These equal the Wick-pairing masses for GUE (2 pairings) and GOE (3 pairings); the extra pairing exists precisely when $-1 \in H$, i.e. iff k is even.

Proof. Decompose by structural type rather than by the value of $S := h_1 + h_3 = h_2 + h_4$.

Step 1 (two universal families). Tuples of type (a): $h_1 = h_2 =: t$, $h_3 = h_4 =: u$ satisfy both equations of (1) identically; there are k^2. Type (b): $h_1 = h_4 =: t$, $h_2 = h_3 =: u$ gives another k^2. Their overlap is $t = u$, of size k, so $|(\mathrm{a}) \cup (\mathrm{b})| = 2k^2 - k$.

Step 2 (no further $S \neq 0$ solutions). If $S \neq 0$, combining the reciprocal equation over a common denominator gives $(h_1h_3 - h_2h_4)\, S = 0$, hence $h_1h_3 = h_2h_4$: both unordered pairs are root pairs of $X^2 - SX + h_1h_3$, which has at most two roots, so the pairs coincide and the tuple is of type (a) or (b).

Step 3 (the $S = 0$ layer). If $S = 0$ then $h_3 = -h_1$, $h_4 = -h_2$, and the second equation holds identically. Such tuples exist iff $-1 \in H$, i.e. iff k is even, and then number k^2, freely parametrised by (h_1, h_2). Within them, $h_2 = h_1$ gives type (a) and $h_2 = -h_1$ gives type (b); these subfamilies are disjoint of size k each, leaving $k^2 - 2k$ genuinely new solutions.

Total. Odd k: no $S = 0$ layer, so $2k^2 - k$. Even k: $(2k^2 - k) + (k^2 - 2k) = 3k(k-1)$. □

The theorem was verified exhaustively on small cases (e.g. $p = 5$, $H = \{\pm 1\}$: six solutions, decomposing as {ab:2, a:2, b:2}) and against brute-force enumeration for many (p, k); see also Experiment 13's artefact discipline.

Remark 3 (Arithmetic shadow)**.** *The third pairing is governed by the unique order-two element -1 of the cyclic group $\mathbb{F}_p^\times$ lying in H: a shadow of $\mathbb{F}_p^\times[2]$, not of any classical symmetry.*

Remark 4 (Wick dictionary, structural)**.** *The three solution types map onto the three Wick pairings of four slots:*

$$\textit{type (a)}\ \big(h_1{=}h_2,\ h_3{=}h_4\big) \leftrightarrow (12)(34), \qquad \textit{type (b)}\ \big(h_1{=}h_4,\ h_2{=}h_3\big) \leftrightarrow (14)(23),$$

$$S = 0\ \textit{layer}\ \big(h_3{=}-h_1,\ h_4{=}-h_2\big) \leftrightarrow (13)(24),$$

the last existing precisely when $-1 \in H$, which is exactly when the count jumps to the GOE-type mass $3k(k-1)$. We stress this as a structural analogy*: the finite-k expressions $2k^2-k$ and $3k(k-1)$ carry endpoint corrections relative to naive Gaussian Wick masses $2k^2$ and $3k^2$, and no Gaussian integral is being evaluated.*

Remark 5 (Reversibility does not force even k)**.** *Call A reversible if some $R \in \mathrm{GL}(2,\mathbb{Z})$ satisfies $RAR^{-1} = A^{-1}$; then a traceless reverser exists and the quantised propagator admits an anti-unitary Θ with $\Theta U_A \Theta^{-1} = U_A^{-1}$. Nevertheless numerous reversible maps admit split primes with odd k: for $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ and $A = \left(\begin{smallmatrix} 5 & 2 \\ 2 & 1 \end{smallmatrix}\right)$, instances include $(p,k) = (101,25), (199,11), (211,21), (199,9)$, where Theorem 2 forces the GUE-type value regardless of Θ. Classical reversibility and multiplicative parity are independent.*

Remark 6 (Degenerate chart: $M_{21} \equiv 0 \pmod N$)**.** *The Weil-cocycle identity $c(M, M^{-1}) = \pm 1$ for $\Theta^2 = U_M\, \overline{U_M} = c(M, M^{-1}) \cdot 1$ is proved for reversing involutions $R = [\![p, q], [\,r, -p\,]\!] \in \mathrm{GL}(2,\mathbb{Z})$ with $M = S \cdot R$ and $\gcd(M_{21}, N) = 1$. The standard symplectic reflection is $S = [\![1, 0], [0, -1]\!]$. The hypothesis is automatic for $R = [\![0, -1], [1, 0]\!]$ and $R = [\![\pm 1, 0], [0, \mp 1]\!]$; it* fails *for $R = [\![1, -1], [0, -1]\!]$ ($M_{21} = 0$) and similar degenerate charts. In that case the twisted Weil cocycle collapses: the M-quantum propagator $U_M(x, y) = (1/\sqrt{N}) \exp(2\pi i (M_{11}xy + M_{12}x^2/2 + M_{22}y^2/2)/N)$ has $M_{21} \equiv 0$, so the (x, y) cross-term vanishes and U_M factorises as* $\mathrm{chirp}x \otimes \mathrm{chirp}y$*. Complex conjugation sends $U_M \mapsto \overline{U_M}$ and $\overline{U_M} \circ U_M = (1/N) \cdot 1$ is the* pointwise *identity on the basis $T(v)$; Θ^2 acts as $+1$ on every basis vector. The $\beta = 1/\beta = 4$ dichotomy collapses trivially to $\beta = 1$. This case is covered by the red-team review [?] and the analytic paragraph in the LEDGER R12-A area; the analytic–empiric gap it exposed is closed by the explicit hypothesis $\gcd(M_{21}, N) = 1$ now stated.*

4 The unmatched sector

4.1 Reduction to H^3

Fix signs $(+,-,+,-)$ and labels $(a;b) \in (\mathbb{F}_p^\times)^8$. Substituting $h_i = h_1 u_i$ ($u_1 = 1$) factors out the scale $h_1 \in H$; each reduced solution $(u_2, u_3, u_4) \in H^3$ lifts to exactly k full solutions, so

$$\mathrm{sol}(a,b) = k\, N_3(a,b).$$

Eliminating u_4 yields one polynomial of bidegree $(2,2)$ in (u_2, u_3), whence the unconditional Schwartz–Zippel bound $N_3 \le 4k$. Empirically the truth is sharper and two-tiered: a small mean with Poisson-like fluctuations, and a tight worst-case ceiling.

4.2 Tail: the tight resonance bound

Call a label pattern $(a;b)$ *generic* when neither slot-pair form is proportional — $a_1 b_3 \neq a_3 b_1$ and $a_2 b_4 \neq a_4 b_2$ — and, in addition, no cross-proportionality $a_i b_j = a_j b_i$ holds for $(i,j) \in \{(1,2),(1,4),(2,3),(3,4)\}$; these are precisely the conditions under which the bidegree-$(2,2)$ elimination polynomial has nonvanishing quadratic part. The proportionality locus is cut out by finitely many nonzero linear equations on the label torus, hence has relative measure $O(1/p)$ in $(G)^8$.

Degenerate (proportionality-containing) patterns violate bounds of the shape below but resolve into finitely many families with their own exact formulae: fully matched labels give Theorem 2; slot-proportional pairs collapse one resonance equation onto the other and contribute $\Theta(k^3)$-type layers; each such family is a lower-dimensional stratum handled separately.

Conjecture 7 (Tight tail bound)**.** *For every generic unmatched pattern,*

$$N_3(a,b) \;\le\; \left\lfloor k^2/p \right\rfloor + 2.$$

The additive slack 2 is not arbitrary: when k^2/p sits just below an integer, the floor drops by one while the arithmetic count does not, so any floor-based law needs unit slack; the observed maximum requires exactly two.

Conjecture 8 (Unified tail bound)**.** *In fact genericity may be unnecessary: for* every *label pattern that is not Wick-compatible — i.e. admits no perfect matching of the four slots into pairs (i,j) with $a_i = a_j$ and $b_i = b_j$ — the reduced count obeys*

$$N_3(a,b) \;\le\; \left\lfloor k^2/p \right\rfloor + 2.$$

Patterns inside the Wick locus carry the exact pairing masses (Theorem 2 and its multi-frequency analogue), so this one bound would subsume the degenerate-strata estimate used by Corollary 14.

Remark 9 (Empirical support for Conjecture 8)**.** *A stratified scan (`review/scripts/degenerate_strata_scan.p` artifacts `audit_degenerate_strata.json`) grew the ladder $p \in \{1009, \dots, 8009\}$ with $k \sim \sqrt{\rho p}$ for $\rho \in \{1, 4, 16\}$ (k up to 364) and compared five non-Wick degenerate classes — slot-proportionality, single cross-, double cross-, chained proportionalities, and the sharpest near-Wick edge where three of the four Wick equalities hold — against a fully generic baseline,* 250 *random instances per cell. All non-Wick classes tracked the generic statistics throughout: maxima stayed within $\{0, \dots, 10\}$ versus the ceiling $\sim\rho{+}2$, means matched the exact ensemble law k^3/p^2, while the absorbed Wick sector gave exactly $\Theta(k)$. No violation or excess over the generic baseline was observed in any class.*

Remark 10 (Smooth-law conditional on Ostafe–Shparlinski–Voloch)**.** *The two conjectures above are unconditional on the empirical record (∼ 293 tested configurations, ∼ 60,000 random label patterns, 0 violations; new lattice $p \le 2003$, $k \le 30$, 800 trials each, reported in* `artifacts/audit_conj_tight_extend`*. A* conditional ***proof*** *of the smooth tail law $N_3 \le k^2/p + \sqrt{k}$ at unit constant is available for $k \ge p^{3/7+\varepsilon}$ by combining the pointwise subgroup Kloosterman bound of Ostafe, Shparlinski and Voloch [6] (Corollary 2.9) with the nested-completion program of* `drafts/tail_bound_attempt.md` *§2. Below the $p^{3/7}$ threshold the only available power-saving-in-p estimate is Bourgain's 2005 JAMS estimate, whose dependence $\delta(\varepsilon)$ is not explicit (per OSV's own Remark 2.11); a published explicit pointwise $\delta > 0$ for $k < p^{3/7}$ is not available, and the precise gap is documented in* `review/subgroup_kloosterman.md` *and* `drafts/tail_bound_proof.md`*.*

Across ∼230 tested configurations ($p \le 2729$, $3 \le k \le 682$, hundreds of random patterns each) there are zero violations; an earlier guess with constant 1 failed on boundary cases where k^2/p sits just below an integer, and the slack of 2 accommodates that granularity.

4.3 Mean: an exact ensemble law

Worst-case and typical behaviour must not be conflated. The typical behaviour is governed by an exactly provable law.

Lemma 11 (Zero counts of linear forms)**.** *Let p be prime, $G = \mathbb{F}_p^\times$, and $c_1, \dots, c_n \in G$. Then the number of solutions in G^n of $c_1x_1 + \cdots + c_nx_n = 0$ equals*

$$Z_2 = p - 1, \qquad Z_3 = (p-1)(p-2), \qquad Z_4 = (p-1)(p^2 - 3p + 3),$$

independently of the coefficients.

Proof. $n = 2$: each $x_1 \in G$ determines $x_2 = -c_1x_1/c_2 \in G$. $n = 3$: fix $(x_1, x_2) \in G^2$; x_3 is admissible iff $c_1x_1 + c_2x_2 \ne 0$, and the excluded pairs are counted by Z_2. $n = 4$: fix (x_1, x_2, x_3); the excluded triples are counted by Z_3. □

Theorem 12 (Ensemble-mean resonance law)**.** *Fix any subgroup $H \le \mathbb{F}_p^\times$, $|H| = k$. With $N_3(a, b)$ as above,*

$$\sum_{(a,b)\in(G)^8} N_3(a,b) = k^3\big[(p-1)(p^2-3p+3)\big]^2, \qquad \langle N_3\rangle = \frac{k^3(p^2-3p+3)^2}{(p-1)^6} = \frac{k^3}{p^2}\Big(1 + O\big(\tfrac{1}{p}\big)\Big).$$

Proof. Swap summations: $\sum_{(a,b)} N_3(a,b) = \sum_{u\in H^3} \#\{(a,b) : A(u;a) = 0,\ B(u;b) = 0\}$ with $A = a_1 - a_2u_2 + a_3u_3 - a_4u_4$ and $B = b_1 - b_2/u_2 + b_3/u_3 - b_4/u_4$. All eight coefficients appearing are nonzero because $u_i \in H \subseteq G$, the two equations involve disjoint variables, and Lemma 11 makes each factor Z_4 independent of u. Multiply by k^3 triples and divide by $(p-1)^8$ labels. □

Experiment 13 (Exact verification)**.** *Monte-Carlo over random labels agrees with Theorem 12 within statistical errors in five configurations $(p, k) = (101, 10), (101, 20), (211, 15), (211, 21), (401, 20)$, up to 40 000 trials each, all $|z| < 1.5$.*

(p, k)	Monte-Carlo mean	exact	trials
$(101, 10)$	0.09573	0.09803	40 000
$(101, 20)$	0.78065	0.78424	20 000
$(211, 15)$	0.07710	0.07581	20 000
$(211, 21)$	0.21127	0.20801	15 000
$(401, 20)$	0.04733	0.04975	15 000

Figure 1: Verification of Theorem 12.

4.4 Complete picture of the fourth moment

Corollary 14 (Matched sector dominates; conditional on Conjecture 7)**.** *Fix an observable with finite frequency support F, $|F| = c_F$. Assume the tail bound of Conjecture 7. Then for $k < p$ and any eigenspace,*

$$\left|\mathrm{Tr}\big(W^4\big) - \big(\textit{matched-sector contribution}\big)\right| \;\le\; C_F\, k\big(k^2/p + 1\big),$$

where C_F depends only on F and $\max|c_n|$; since the matched sector contributes $\Theta(k^2)$ (Theorem 2), the relative unmatched contribution is $O_F(k/p+1/k) \to 0$ as $p \to \infty$ with $k = o(p)$. More generally, any *uniform bound of the shape $N_3 = O(k^\theta)$ with $\theta < 1$ per pattern yields the same conclusion; establishing such a sublinear bound unconditionally is Open Problem 21.*

Proof. There are at most $|F|^4 = O_F(1)$ signed label patterns. By the master-formula bridge of Section 2, each pattern's moment contribution is bounded in absolute value by its count times $\max|c_n|^4$ (phases have unit modulus). Generic patterns satisfy $\mathrm{sol}(a, b) = kN_3(a, b)$ with $N_3 \le \lfloor k^2/p \rfloor + 2$ under the conjecture; degenerate patterns belong to finitely many strata whose total is likewise $O_F(k^3/p + k)$ using their explicit forms. Summing over patterns gives the displayed bound. □

Remark 15. *The degenerate-stratum estimate invoked in the proof of Corollary 14 ("explicit forms") is not written out in this paper: it requires classifying the finitely many proportionality types of label patterns and bounding each. The stratified scan of Remark on Conjecture 8 supports a stronger and cleaner replacement — every non-Wick pattern obeys the same ceiling as generic ones, while Wick-compatible patterns are absorbed into the matched sector with their exact masses — which, once proved, closes this obligation completely. All computed instances are consistent with it; see Open Problem 21.*

Combining the three layers: the *matched* sector contributes exactly $\Theta(k^2)$ through Wick pairings (Theorem 2); the *typical* unmatched pattern carries mean k^3/p^2, negligible once $k < p^{2/3}$ (Theorem 12); the *worst-case* unmatched pattern is bounded *conjecturally* at $\lfloor k^2/p \rfloor + 2$ (Conjecture 7, supported by $0/\sim 293$ violations across $\sim 60{,}000$ random patterns; see Remark 10). Under this conjecture, the fourth-moment sum is dominated by Wick pairings for all $k \ll p$ (Corollary 14).

Scope limitation. Experiments at larger eigenspace dimensions (m up to 69) show that higher moments $\mathrm{Tr}(W^{2r})$, $r \ge 3$, are dominated by a few large matrix elements spanning orders of magnitude — an arithmetic hierarchy incompatible with naive universality. Raw compressions do not by themselves deliver local spacing laws; some symmetrisation appears necessary (cf. Section 6).

5 The sixth moment: strata laws and an arithmetic excess

At $r = 6$ with signs $(+,-,+,-,+,-)$ every solution tuple partitions $\{1,\dots,6\}$ by equal values. Brute-force enumeration (sign-checked; an earlier inversion of the right-hand sums produced invalid tables, now retracted) separates purely combinatorial strata from arithmetic ones.

Proposition 16 (Exact strata identities)**.** *For any $H \le \mathbb{F}_p^\times$, $|H| = k$:*

$$[6] \;=\; k, \qquad [2,4] \;=\; 9k(k-1) \quad \textit{(both parities)},$$

and, for odd *k,* $[2,2,2]_{\text{slotwise}} \;=\; 6k(k-1)(k-2)$*. Here $[\cdot]$ counts tuples whose value-equality pattern has the displayed block sizes. Proofs mirror Theorem 2: e.g. in $[2,4]$ resonance forces each value to occupy equally many $+$ and $-$ slots, giving 3×3 slot choices times $k(k-1)$ ordered value pairs.*

Conjecture 17 (Combinatorial skeleton)**.** *When the arithmetic strata vanish (observed at $(p,k) = (101,10),(211,10),(211,14),(401,16),(101,20)$),*

$$\mathrm{sol}_{+-+-+-}(H) \;=\; 5k\,(3k^2 - 9k + 8),$$

with all remaining even-k strata carrying the fixed forms $[1,1,1,1,2]$ and $[2,2,2]_{\text{even}}$ recorded numerically in the artefact tables.

The skeleton is *not* the whole story: at $(p,k) = (241,30)$ enumeration gives $\mathrm{sol}_6 = 417\,900$ against the skeleton value $365\,700$. The excess decomposes exactly into arithmetic strata absent in the small cases,

$$[1^\times 6] = 50\,760, \quad [1,1,2,2] = 1\,080, \quad [1,1,1,3]_{\text{excess}} = 360,$$

so the general even-k law reads

$$\mathrm{sol}_6 \;=\; 5k(3k^2 - 9k + 8) \;+\; A_p(k), \qquad A_p(k) \ge 0,$$

where $A_p(k)$ aggregates Kloosterman-type coincidences among disjoint triples (same quantity $D_p(k)$ governing the odd case). Characterising $A_p(k)$ is open (cf. Open Problem 21); its vanishing locus at least includes the five configurations above.

Proof of the strata identities. Partition strata are labelled by block-size multisets. *Stratum* $[6]$: all six values equal, any $h \in H$: k tuples. *Stratum* $[2,4]$ (values x twice, y four times, $x \ne y$): resonance forces net signed multiplicity zero for each value, so the two x's occupy one $+$ and one $-$ slot while the four y's fill the rest; choosing the $+$-slot of x (3 ways) and its $-$-slot (3 ways) and the ordered pair (x,y) $(k(k-1))$ gives $9k(k-1)$ — matching enumeration at every tested k, both parities. *Stratum* $[2,2,2]$*, odd case*: no sign-pairs exist ($-1 \notin H$), so pairs are equal-value; three ordered distinct pair values give $6k(k-1)(k-2)$, again exact at $k = 15$. The identity $[1,1,1,3] = 18k(k-2)$ holds on the combinatorial skeleton (verified identically at $k \in \{10,14,16,20\}$ across three primes); at $(241,30)$ an arithmetic excess of 360 sits on top of it, consistent with the decomposition below. □

For *odd* k a polynomial part persists — $6k(k-1)(k-2) + 9k(k-1) + k$ from the three proven strata — plus an arithmetic remainder $D_p(k) = A_p(k)$ counting disjoint-triple pairs (T_+, T_-) with equal sums and reciprocal sums. Both are genuinely p-dependent: at $k = 25$, D_p takes different values for $p = 101$ versus $p = 401$ (14 400 vs 3 600), and the sweep over nine configurations gives $D_p(k)\,/\,(k^6/p^2) \in [0.53,\ 4.04]$ with no collapse — Kloosterman-type oscillation, the same wall met in Open Problem 21.

6 Perturbation-induced universality

The resonance core controls one moment; local statistics require resolving the arithmetic degeneracies.

Lemma 18 (Perturbation splitting)**.** *Let E be drawn from the Gaussian unitary ensemble* GUE(N)*: $E = E^\dagger$ Hermitian with density proportional to $e^{-\operatorname{tr} E^2}$ — equivalently $E = (X + X^\dagger)/2$ with X an i.i.d. complex Gaussian matrix — and consider the* unitary *perturbation flow*

$$U_\varepsilon \;=\; e^{i\varepsilon E}\, U_A, \qquad \varepsilon > 0.$$

Assume the spectral gap $\delta_* := \min_{\lambda' \neq \lambda} |\lambda' - \lambda|$ of U_A satisfies $\varepsilon\, \|E\| \ll \delta_*$, so that distinct multiplets do not mix through second order. Then, for each level $\lambda_j = e^{i\theta_j}$ inside the λ-multiplet,

$$\theta_j(U_\varepsilon) \;=\; \theta_j + \varepsilon\, \mu_j \;+\; O\big(\varepsilon^2 \|E\|^2/\delta_*\big),$$

where $\mu_1, \ldots, \mu_m$ are the eigenvalues of the compression $P_\lambda E P_\lambda$. By unitary invariance of the GUE law this compression is distributed exactly as an $m \times m$ GUE(m) *matrix. Hence, as $m \to \infty$ at fixed $\varepsilon\|E\|/\delta_* \to 0$, unfolded intra-multiplet spacings of U_ε converge to the spacing law of the GUE (the Wigner surmise serving as its standard numerical proxy).*

Proof sketch. Write $V(\varepsilon)$ for the unitary group path; Kato's first-order degenerate perturbation theory gives the phase shifts $\varepsilon\mu_j$ as eigenvalues of the compression, with the stated second-order remainder controlled by the gap δ_*. Unitary invariance of GUE under conjugation makes $W^\dagger E W \overset{d}{=} E$ for any fixed unitary W; compressing an invariant law onto any fixed m-dimensional subspace yields the GUE(m) law. Standard GUE spacing asymptotics complete the claim. A purely additive model $U_A + \varepsilon E$ gives the same first-order splittings (the two models differ by conjugation to first order), which is why either may be used numerically; we use the unitary one throughout. □

Experiment 19 (Multi-prime confirmation)**.** *At primes with small quantum period $p \in \{1597, 2161, 2207, 3001\}$ (multiplicities $m \approx 47$–69) we implement the unitary flow of Lemma 18 with $\varepsilon = 10^{-4}$, track eigenvalues across the flow by optimal (Hungarian) assignment on $|\lambda^{new} - \lambda^{old}|$, unfold each multiplet to unit mean spacing, and evaluate the Kolmogorov–Smirnov statistic* per prime *on the pooled spacings of its three largest multiplets: Appendix A specifies every procedural step. (For the Arnold map two of these four primes are inert, $p \in \{1597, 2207\}$; the pipeline and Lemma 18 do not use splitting, so the experiment covers both arithmetic types.)*

p	*spacings*	KS *vs WD*	KS *vs Poisson*	*verdict*
2207	*204*	*0.0638*	*0.2778*	*WD*
1597	*138*	*0.0672*	*0.2828*	*WD*
2161	*160*	*0.0689*	*0.2808*	*WD*
3001	*178*	*0.0756*	*0.2911*	*WD*

All four statistics lie below the 5%-level critical value ≈0.096 for $n \approx 200$, i.e. every prime individually is consistent with Wigner–Dyson, while Poisson is rejected outright everywhere. An ε-sensitivity sweep ($10^{-3}, 10^{-4}, 10^{-5}$ at $p = 2207$) keeps KS_{WD} in $[0.064, 0.091]$ with no drift, indicating the measurement sits in the first-order regime of Lemma 18. WD here means the pooled surmise test against the Wigner–Dyson family; at these sample sizes GOE and GUE surmises are statistically indistinguishable (critical value ≈0.096 for $n \approx 200$), so we do not discriminate $\beta = 1$ vs $\beta = 2$. Poisson is clearly rejected throughout.

Experiment 20 (Beta resolution at scale)**.** *At three further primes with small quantum period* — $(p, K, m) \in \{(3571, 17, 210), (4481, 32, 70), (5779, 27, 214)\}$ — *the same unitary pipeline yields* $n = 13\,723$ *pooled spacings. At such sizes the analytic Wigner* surmises *carry a fixed systematic offset from the exact spacing laws (*$\sim$*0.05 in KS), so discrimination is made against Monte-Carlo reference distributions of* GOE(m) *and* GUE(m) *built at the very multiplet dimensions observed (*$397\,800$ *reference spacings each). Result:*

$$\mathrm{KS}_{\mathrm{data}\,\|\,\mathrm{GOE}} = 0.0676, \qquad \mathrm{KS}_{\mathrm{data}\,\|\,\mathrm{GUE}} = 0.0099, \qquad \mathit{gap} = 0.0577 \gg 0.0118 = \mathit{crit}_{5\%}.$$

The data are statistically indistinguishable from GUE and clearly exclude GOE — precisely the prediction of Lemma 18, since the perturbation ensemble is GUE.

7 What remains open

Open Problem 21 (Prove the tail bound)**.** *Establish Conjecture 7 with explicit constant. Schwartz–Zippel gives* $4k$*; the empirical ceiling is* $\lfloor k^2/p \rfloor + 2$*. Likely input: Weil-type bounds for curves intersected with multiplicative subgroups, or sheaf monodromy.*

Open Problem 22 (Sixth and higher moments)**.** *Characterise the arithmetic excess* $A_p(k)$ *in Section 5: its vanishing locus, its size relative to* k^6/p^2*, and the corresponding closed strata forms. Extend the matched/unmatched split uniformly in even* $r \geq 8$*.*

Open Problem 23 (Unconditional local law)**.** *Replace Gaussian randomness in Lemma 18 by arithmetic perturbations, and bridge from splitting statistics to a nearest-neighbour law for the unperturbed compressions. A direct attempt already* fails*: driving the flow with a Hermitian Weyl combination* V_{arith} *(*$U_\varepsilon = e^{i\varepsilon V_{\mathrm{arith}}} U_A$*,* $\varepsilon = 10^{-4}$*; primes* 2207, 4481*) produces intra-multiplet spacings with* KS ≥ 0.15 *against* both *Monte-Carlo GOE and GUE references (*$n = 2175$, 4417*) — far outside the random-matrix class and consistent with the heavy-tailed singular-value hierarchy of the compression* $P_\lambda V_{\mathrm{arith}} P_\lambda$ *(eigenvalue excess kurtosis* 5.9 *and* 9.3, $\mathrm{ev}_{\max}/\sigma = 3.8$ *and* 5.0*; artifact* `taskA_entry_gaussianity.json`*). (eigenvalue excess kurtosis* 5.9 *and* 9.3, $\mathrm{ev}_{\max}/\sigma = 3.8$ *and* 5.0*; artifact* `taskA_entry_gaussianity.json`*). Unconditional local law therefore requires genuine symmetrisation (Hecke-type averaging or equivalent), not merely a deterministic replacement for the random input. A finer diagnosis narrows the target further: at the split primes* 4481 *(*$m = 71$*) and* 3571 *(*$m = 211$*) the* entries *of* $P_\lambda V_{\mathrm{arith}} P_\lambda$ *are individually consistent with Gaussians (excess kurtosis* $|{<}0.15|$ *off diagonal, KS p-values* >0.25*), while their eigenvalues show strong outliers (excess kurtosis* 5.9 *and* 9.3*;* $|\lambda|_{\max}/\mathrm{std} \approx 3.8, 5.0$*). The failure of random-matrix behaviour is thus driven by entry* correlations*, not by marginal non-Gaussianity: Kurlberg–Rudnick-style Gaussianity of matrix elements already holds inside raw multiplets, and what is missing is decorrelation. A first Hecke-symmetrisation test (*`src/hecke_local_law_test.py`*, draft* `drafts/hecke_symmetrization_test.md`*,* 8 *configs,* $m \leq 17$*,* 32 *Haar-surrogate rotations) confirms the diagnosis: off-diagonal kurtosis collapses from raw* $+5.6$ *to Hecke* -0.7 *in the worst case, and the eigenvalue spread drops modestly (*$2.9 \to 2.6$*). The Hecke-averaged compression is therefore* strictly better-behaved *than the raw arithmetic perturbation (R49,* $\mathrm{KS}_{\mathrm{GUE}} = 0.20$–$0.28$*), but does not at* $m \leq 17$ *deliver Wigner spacing (*$\mathrm{KS}_{\mathrm{GOE}} \in [0.13, 0.35]$*, inconclusive at this* m*). Closing the gap requires a Hecke basis of the actual centraliser torus (not the Haar surrogate) and a larger* $m \geq 70$ *test.*

Open Problem 24 (Inert primes and general N)**.** *Extend to* Δ *a non-residue, where* $\alpha \in \mathbb{F}_{p^2} \setminus \mathbb{F}_p$*, and to composite* N*.*

8 Summary of statuses

Claim	Status
Collision lemma $C(u,v) \le 4$	PROVEN (Lean-formalise
U1 fibre ≤ 2 (Rem. 1)	PROVEN on the *non-deg*
U4 conditional fine-scale	PROVEN conditional, RE
T13 $\Theta^2 = +1$ (Rem. 5 + Rem. 6)	PROVEN UNCONDITIO
r=5 master formula (`src/master_general.py`, `artifacts/master_general.json`)	NUMERICALLY-SUPPO
T21 unconditional local law at the mean-spacing scale	OPEN GAP; the publishe
Matched identity (Thm. 2)	PROVEN
Zero-count lemmas (Lem. 11)	PROVEN
Ensemble-mean law (Thm. 12)	PROVEN; verified 5/5
Reversibility $\not\Rightarrow$ even k (Rem. 5)	verified, elementary
Tail bound (Conj. 7)	NUMERICALLY-SUPPO
Tail bound below $p^{3/7}$	OPEN (named, OSV Rem
Hecke-centred entry decorrelation	RETRACTED 2026-08-28
Domination, Cor. 14	PROVEN under Conj. 7.
Perturbation splitting (Lem. 18)	PROVEN (sketch)
WD-not-Poisson under unitary flow (Exp. 19)	NUMERICALLY-SUPPO
Beta class = GUE under GUE flow (Exp. 20)	RESOLVED (gap 5× crit
GOE-vs-GUE discrimination	indistinguishable at curre
Unconditional local law	OPEN GAP
Full BGS for raw compressions	NOT CLAIMED

A Methodological details

A.1 Random-label ensembles (Sections 4)

For a fixed (p,k), H is generated as the orbit of $\alpha = g^{(p-1)/k} \bmod p$ with g a primitive root, verified by checking $\mathrm{ord}(\alpha) = k$. Label patterns are drawn i.i.d. uniform on $\{1,\dots,p-1\}^8$ independently per trial; trials-per-configuration range from 300 (tail-bound scan) to 40 000 (mean-law verification), as recorded per experiment. The generic/nondegenerate filter of Section 4 was *not* applied in post-processing; since proportionality events have probability $O(1/p)$ each, their expected count in any scan is $\ll 1$, so violations-by-degeneracy cannot explain the observed zero violation rates.

A.2 Mean-law Monte Carlo

Each trial draws one label pattern and computes N_3 exactly by meet-in-the-middle over $(u_2,u_3) \in H^2$ with u_4 determined by the first equation and screened for membership in H before testing the second equation. Reported z-scores use the exact mean of Theorem 12 as centre and the Poisson-scale standard error $\sqrt{\langle N_3\rangle/\mathrm{trials}}$, consistent with the near-Poisson fluctuation of the count.

A.3 Perturbation pipeline

For each prime p: (i) diagonalise U_A, cluster eigenvalues at tolerance 10^{-4} on rounded real/imaginary parts; (ii) draw $E \sim \mathrm{GUE}(N)$ and realise $e^{i\varepsilon E}$ through its own eigendecomposition; (iii) match perturbed to unperturbed levels by the Hungarian algorithm on the cost matrix $C_{ij} = |\lambda_i^{\mathrm{new}} - \lambda_j^{\mathrm{old}}|$;

the maximum assignment distance scales linearly in ε (7.6×10^{-8} at $\varepsilon = 10^{-5}$, 7.6×10^{-7} at 10^{-4}), confirming first-order tracking; (iv) group matched levels by their origin multiplet, take the three largest multiplets, unfold intra-multiplet gaps to unit mean spacing, pool them within the prime; (v) evaluate KS once per prime against the GOE Wigner surmise CDF $1-e^{-\pi s^2/4}$ and against Poisson $1 - e^{-s}$; no cross-prime pooling is performed. Cluster sizes were stable across $\varepsilon \in [10^{-5}, 10^{-3}]$ (e.g. 69×5 at $p = 2207$).

A.4 Reproducibility

All scripts live in the repository with fixed seeds recorded in their artefact headers; the artefact manifest maps every number above to script and command. Compiled artefacts: `unitary_perturbation.json`, `run46_exact_mean.json`, `run36_audit.json`.

References

[1] O. Bohigas, M.-J. Giannoni, C. Schmit, *Characterization of chaotic quantum spectra and universality of level fluctuation laws*, Phys. Rev. Lett. **52** (1984), 1–4.

[2] J. H. Hannay, M. V. Berry, *Quantization of linear maps on a torus — Fresnel diffraction by a periodic grating*, Physica D **1** (1980), 267–290.

[3] P. Kurlberg, Z. Rudnick, *Hecke theory and equidistribution for the quantization of linear maps of the torus*, Duke Math. J. **103** (2000), no. 1, 47–77.

[4] P. Kurlberg, Z. Rudnick, *On quantum ergodicity for linear maps of the torus*, Comm. Math. Phys. **222** (2001), 201–227.

[5] F. Faure, S. Nonnenmacher, S. De Bièvre, *Scarred eigenstates for quantum cat maps of minimal periods*, Comm. Math. Phys. **239** (2003), 449–492.

[6] A. Ostafe, I. E. Shparlinski, J. F. Voloch, *Equations and character sums with matrix powers, Kloosterman sums over small subgroups and quantum ergodicity*, preprint, arXiv:2110.10941v2 [math.NT] (2021).